EXTRAI: correcting and creating meta-analyses with local language models — a pre-registered five-phase benchmark with a perturbation reading proof and a self-correcting answer key

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Abstract

Language models are entering systematic-review workflows, but most evaluations use cloud models, cannot separate reading from training-data recall, and treat the human answer key as infallible. EXTRAI is a pre-registered, five-phase benchmark of five quantized open-weight models (8–14.8B parameters, one consumer mini-PC, integrated GPU, no cloud calls) as meta-analysis workers over two published meta-analyses and their 21 randomized trials, under two controls most evaluations lack: a *perturbation reading proof* (load-bearing numbers displaced under sealed maps, so reciting a published value is detectable) and a *two-layer, source-verified answer key* with a quotation-backed adjudication rite that puts the human key itself on trial — accumulating a seventeen-entry public errata file against one published anchor (eleven source-confirmed errors, definitional divergences, and the adjudicators' own logged mistakes), and correcting its own cells twice. The five phases ascend from reading to deployment. Reading: 140 perturbed sheets, zero attributable recitations, replicate stability 77–99%. Arithmetic: 1 of 57 unaided confidence intervals exact, rising to 82% through a trivial text-protocol calculator; pooled calls stayed beyond the whole size class. Creation: *all five* models' sheets reproduced the published dichotomous pools (morbidity 0.771–0.779 vs. 0.778), while the continuous diamond separated them — exactly one model, `gemma4:12b`, landed beside the published -0.24 $[-0.32, -0.16]$, at -0.27; the rest landed -0.47 to -0.80 for named reading failures — never arithmetic, and never fabrication: an invention screen across both corpora found zero fabricated values in any model. Orchestration: under a warn-only ten-net detection harness, warnings were resolved by confirm-or-correct and the model's own pooling call closed digit-consistent with its executed results. Deployment, on clean texts with no answer key in any loop: the model sided with the primary sources at all seven source-confirmed errata of the published meta-analysis; its one failed tolerance decomposed to a single mislabeled-dispersion cell seeded by the source's own two-layer notation; and a three-configuration comparison showed conversational and sheet-level detection nets are complements, not substitutes. A built-in robustness check — the instruments were drafted in Portuguese and frozen in English, with the development record archived — found every phase's result within replicate-level variance across the language change, the flagship diamond identical in point estimate and interval (-0.27 $[-0.38, -0.17]$ in both), so none of the conclusions is an artifact of instruction language. All protocols, amendments, seals, sheets, graders, adjudications and raw outputs are public and archived.

1 Introduction

Systematic reviews and meta-analyses are the reference standard of clinical evidence synthesis, and they are labor-bound: extracting structured data from primary trials, checking it, and

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computing pooled effects consume the majority of a review team's hours. Large language models are entering exactly this workflow. Feasibility studies report promising but uneven accuracy for trial data extraction [Schmidt et al., 2024]; assisted extraction has proved noninferior to human-only extraction inside a live review [Lee et al., 2026]; screening assistance is approaching routine use [Xie et al., 2026]; and end-to-end platforms now report complete systematic reviews produced in hours [Bakker et al., 2026], with fully automated drafts rated above a published human-authored review by blinded specialists [McLaughlin et al., 2026]. The major evidence-synthesis organizations have jointly conditioned such use on demonstrated rigor, human oversight and transparent reporting [Flemyng et al., 2025].

Three gaps run through this literature. First, most evaluations use commercial cloud models, while clinical text pushes in the opposite direction: locally run open-weight models keep sensitive material on-premises and have shown extraction performance approaching expert annotation [Wiest et al., 2024, Spaanderman et al., 2025]. Second, benchmarks that score agreement with a published review cannot distinguish a model that *read* the primary trial from one that *remembers* the review — both the anchor and its primaries are plausibly in training corpora, and contamination inflates apparent competence [Xu et al., 2024]. Third, the published review itself is usually treated as an infallible answer key, although hand-made evidence tables carry their own errors; a benchmark that cannot notice when the published value is the one that is wrong will misgrade its best readers.

EXTRAI was designed to close those three gaps at once: quantized open-weight models on a consumer machine with no cloud calls; a perturbation reading proof, in which load-bearing numbers are displaced in every source text under sealed maps so that recitation of published values is detectable; and a two-layer, source-verified answer key with a quotation-backed adjudication rite that puts the human key itself on trial — a verification process that accumulated a seventeen-entry public errata file against one published anchor — eleven source-confirmed errors among them — while logging the adjudicators' own mistakes in the same file [Barra, 2026]. Because the instruments were drafted in the research team's working language before being frozen in English, the study additionally carries a built-in robustness check on instruction language, reported with the results.

In this study, we conducted a single pre-registered, five-phase benchmark of local language models as meta-analysis workers — reading, arithmetic, creation of two published meta-analyses, orchestration under a warn-only detection harness, and deployment on clean texts with no answer key in any loop — comparing five integrated-GPU models under contamination-controlled, source-verified grading against the published values.

2 Methods

2.1 Study design and pre-registration

This was a pre-registered, five-phase benchmark of five local language models as meta-analysis workers, graded against two published meta-analyses under a contamination control and a source-verified key. The study protocol — phase order, model cast, hypotheses and compute budget — was registered and committed to the public repository before any run; the two extensions executed within the deployment phase were likewise registered as dated amendments before their runs. Corpora, sealed perturbations, answer keys, deterministic engines, graders and the detection harness were constructed and frozen before the campaign, during an instrument-development phase whose complete record is archived alongside this study's [Barra, 2026]; Section 2.13 describes the robustness analysis that phase additionally enables. All prompts, seals, keys, harness code, graders, raw model outputs, run logs and adjudication records are public.

The campaign comprised five phases in an ascending line of investigation: P1 *Read* (multimodel structured extraction under the perturbation reading proof), P2 *Calculate* (meta-analytic arithmetic unaided versus through a text-protocol calculator), P3 *Create* (deterministic engines assemble both meta-analyses from each model's sheets, with an erratum-aware comparison against the published values), P4 *Orchestrate* (the best-performing reader executes its own calculations under a warn-only detection harness), and P5 *Deploy* (clean, unperturbed texts with no answer key in any loop, including a three-configuration comparison of error-detection architectures).

2.2 Anchor meta-analyses and corpora

Two recently published meta-analyses served as anchors. Anchor 1 addresses goal-directed fluid therapy (GDFT) versus conventional fluid therapy in major surgery [Ashraf et al., 2026]; its 14 primary randomized controlled trials constitute the first corpus (8 open-access, obtained from PubMed Central; 6 paywalled, legally obtained and never redistributed — scripts in the repository rebuild that stratum from a reader's own copies). Anchor 2 addresses low-carbohydrate diets and glycemic control (HbA1c) in type 2 diabetes [Khraise, 2026]; its 7 primary trials constitute the second corpus. Anchor 1 supplies dichotomous outcome families (overall morbidity, mortality, postoperative ileus) and two continuous ones; as in any meta-analysis, each outcome pools only the trials that report it, so the anchor's own outcome tables roster 4, 2 and 3 of the 14 trials for morbidity, mortality and ileus respectively — indeed, no outcome in the anchor is reported by more than 4 of its 14 trials, which meet all together only in its descriptive characteristics and risk-of-bias tables. All 14 trials are extracted here, each contributing to whichever outcome families it reports, and the extraction-level analyses (cell grading, stability, recitation and invention screens, and the per-trial three-way tables) span all 14. Anchor 2 supplies a single continuous outcome (change in HbA1c, %) reported by all 7 of its trials, whose published random-effects pooled estimate, -0.24 [-0.32, -0.16], is the study's external reference value.

2.3 The perturbation reading proof

Before any model reads a trial, a sealed, deterministic operator displaces load-bearing numbers in its text. A model that returns a displaced value read the text in front of it; one that returns the original published value is reciting training data [Xu et al., 2024] — without this control, agreement with a published meta-analysis cannot be distinguished from memory of it. Models operate exclusively in the perturbed world; the original↔perturbed maps are held by the graders alone and applied only after extraction, as a pure table lookup (the "unperturbation lens"). The sealed maps were produced once, before any model run, and reused unchanged across all phases; their SHA-256 fingerprints were printed into each phase's run log. Phase P5 deliberately runs on clean texts and therefore makes no reading-proof claim (Section 2.11).

2.4 Answer keys and the adjudication rite

Each corpus is graded against a two-layer, source-verified answer key constructed during instrument development: layer 1 holds the anchor's published cells; layer 2 holds each cell verified against the primary's full text, with a literal supporting quotation and a per-cell verdict. The key itself is on trial by design — source verification of Anchor 1 accumulated a public errata file of source-confirmed entries against the published meta-analysis (Table 1; the file also keeps the adjudicators' withdrawn accusations, struck through, and its pending items), and the campaign graded against the key as corrected through the most recent adjudications. Anchor 2's verification confirmed no errata: its statistics reproduced cleanly.

Table 1: The source-confirmed errata of Anchor 1, as graded against in this study (confirmed items only; full literal quotations, the withdrawn adjudicator accusations and pending items are in the archived errata file). Conclusion-level impacts are analyzed in the Discussion.

#	Where (anchor)	Published	Source says	Impact
1	Characteristics, Yoon	GDFT 36 / control 39	GDFT n=39; control n=36	none (outcome row was already correct)
3	Characteristics, Weinberg ASA	"Not stated"	I–II 7 (27%); ≥III 19 (73%)	description only
5	Characteristics, Sujatha ASA	"95:105" + a spreadsheet-time–corrupted cell	eligibility only (ASA I–II), no distribution	description only
9	Oral-diet outcome, Sun	72±24 vs 96±30 h	"shorten … by 2 days"; medians 4.0 vs 6.0 d	magnitude ∼2× off, direction unchanged
10	Characteristics, de Waal ASA	GDFT 24:123:86:1	inverse: 123 = 52.6% × 234 (control arm)	description only
11	Characteristics, Diaper ASA	"Not stated"	III–IV 98 (50.0) / 85 (42.9)	description only
12	Flatus outcome, 2 trials	55±14/58±16 and 52±15/60±18 h	"flatus" absent from both full texts	2 of 3 pool rows unverifiable
13	Sample sizes, six trials	presented as randomized	analyzed counts, undeclared	none demonstrated (layer-robust pools)
15	Pooled morbidity caption	"Mantel–Haenszel"	numbers reproduce under DerSimonian–Laird	label only
16	Ileus outcome, Castro	ileus 6 (14.0%) / 19 (45.2%)	"ileus" absent; counts are the trial's PPC result	**conclusion-level (Discussion)**
17	Fluid table, Coeckelenbergh blood loss	GDFT 500 / control 450	the labeled parallel sentences give the inverse	description only

Mechanical grading is backed by an adjudication rite: no cell changes category without the source quotation that decides it, and the adjudicators' own errors are published as errata in the same file as everyone else's. The rite exists because judges exhibit a measurable *plausibility bias*: during key construction, the graded record reversed the benchmark's own human–AI adjudicator six times — including two withdrawn accusations of model fabrication in which the models had extracted literally correct values that rigid verification windows had hidden [Barra, 2026]. Quotation-before-verdict is the structural constraint on that bias, for human, hybrid and model judges alike. The adjudication layer itself is that hybrid: a human–AI pair — the author working with a cloud-based frontier assistant that implements the graders and drafts verdicts under the rite (disclosed in the Acknowledgments) — so the study's local-only property is scoped to the *evaluated pipelines* (Section 2.6), never to the grading and review layer, and every adjudication this layer produced is recorded with its deciding quotation.

2.5 Instruments

All model-facing instruments are frozen English prompts, committed to the study's record at registration: the structured extraction sheets (JSON forms with per-cell `{"value", "where"}` objects), the arithmetic tasks, the calculator protocol, the harness dialogue, and the trigger and synthesis prompts. The instruments were originally drafted in Brazilian Portuguese, the research team's working language, and translated field-by-field under a published correspondence table (for example, `ic95_md`→`ci95_md`; `RESULTADO`→`RESULT`); the originals are archived, and Section 2.13 describes the robustness analysis this enables.

2.6 Models and hardware

Five quantized open-weight models sized for an integrated GPU formed the comparison cast: `gemma4:12b`, `qwen3:14b`, `llama3.1:8b`, `qwen3.5:9b` and `deepseek-r1:14b`, each pinned by its

Ollama build digest in the repository's model manifest. All phases ran on one consumer mini-PC (AMD Ryzen 7, Radeon 780M integrated GPU, 32 GB RAM, no discrete GPU), served by Ollama with a 16,384-token context, reasoning channels disabled, and a uniform 4,000-token output allowance for extraction tasks; no cloud API was called anywhere in any evaluated pipeline (the grading and adjudication layer, by contrast, is human–AI and cloud-assisted; Section 2.4). One model was resident at a time, with an explicit memory unload between models. Phase P4 used only `gemma4:12b`, pre-designated in the protocol as the orchestration candidate on instrument-development evidence.

2.7 Phase P1 — structured extraction

Each of the five models extracted all 14 perturbed Anchor-1 primaries from zero, twice (two independent replicates), under the frozen English extraction sheet (30 fields per trial, each an object `{"value", "where"}` — the mandatory `where` is the model-side provenance instrument: a locator naming the section or table each value was taken from, not a verbatim quotation; these locators are raw data in every archived sheet and were load-bearing in adjudication, e.g. identifying which of three population layers a sample size came from). Grading used the eligible cell set of the two-layer key — cells with a usable, digit-bearing source-verified value; 124 cells per model — comparing the seal-reversed first-parseable sheet against the key's source layer with a magnitude-based numeric comparator, declared an approximation of the full ruler (it cannot see summarization equivalence or population-layer choices); all non-matching cells were listed for adjudication rather than auto-verdicted. Replicate stability was the same comparator applied between the two raw replicates. Recitation candidates were detected mechanically — a cell whose raw text contains a seal pair's original value while the displaced image the model actually read is absent — and each candidate was adjudicated against the perturbed source text before any verdict. Divergent cells were further classified by factuality: *omission* (the model wrote *not reported* where the source-verified key holds a value), *invention candidate* (the cell contains a number absent from the perturbed text and not derivable from it by percentage, sum, difference or mean), or *value/format divergence*; the same invention screen was applied to every cell of the Anchor-2 sheets of phase P3, both replicates.

2.8 Phase P2 — meta-analytic arithmetic

Each model computed, over its own P1 sheets, per-study risk ratios and mean differences with 95% confidence intervals, and pooled estimates, in two arms: arm A unaided (no tool), and arm B through a plain-text calculator protocol in which the model writes lines of the form `CALC: function(arguments)`, the harness executes the validated function and returns `RESULT:` into the context (up to 5 rounds and 20 calls), and the model then answers with a final JSON. Three prompt families (risk ratios, mean differences, pools) $\times$ two arms $\times$ two replicates were run per model. Grading recomputed the ground truth with the same validated engine from the same input block the model saw, under pre-registered parsing rules (event counts read from forms such as "19 (32.8%)"; percent-only cells converted against the group size; means accepted only as mean $\pm$ SD, so that median (IQR) cells have no computable truth and *not computable* is the correct answer there). Each reported quantity was labelled exact, right-direction, wrong, correctly-refused, wrongly-refused, or unsupported; intervals and pooled estimates were graded separately.

2.9 Phase P3 — deterministic creation of both meta-analyses

Everything downstream of the sheets was executed by pre-registered, frozen code; no model touched a number after extraction. For Anchor 1, the engines parsed each model's P1 sheets, selected event/denominator routes per outcome, computed per-study risk ratios and pooled them under both Mantel–Haenszel and DerSimonian–Laird [DerSimonian and Laird, 1986], applied the sealed reversal, and classified every difference from the published tables into a frozen five-category, erratum-aware comparison (*reproduces / differs by a documented anchor erratum / model route / model error / source unavailable*); the pooled comparison is made under DerSimonian–Laird because the published pooled numbers instantiate that method under a Mantel–Haenszel caption (anchor erratum #15). For Anchor 2, each model freshly extracted the 7 perturbed primaries (two replicates) under the English continuous-outcome sheet — which, unlike the Anchor-1 sheet, records bare values with no provenance field, an instrument asymmetry inherited from its development version; a deterministic route selector then built each trial's sextet (mean, SD, n per arm) by the frozen rules — declared change used directly; standard errors and confidence intervals converted to SDs; baseline-and-final pairs combined under the Cochrane $r = 0.5$ rule [Higgins et al., 2023] — and the DerSimonian–Laird pool was computed twice: in the perturbed world, and through the graders' sealed lens for comparison with the published -0.24 $[-0.32, -0.16]$. Comparison tolerances were pre-registered at ± 0.01 for risk ratios and interval bounds and ± 0.1 for mean differences.

2.10 Phase P4 — orchestration under a warn-only detection harness

The study's ten-net detection harness follows the doctrine that nets *detect and warn, never substitute a value.* The doctrine is empirical, not aesthetic: in instrument-development testing, checking stages given write-power did net harm through the same plausibility bias — an auditing model "repaired" correct cells by imputing plausible values (once correcting a sample size to 24.5 while quoting, in its own verdict, the text's "n = 28"), a calculator stage flipped the sign of a control arm that had genuinely worsened (+0.3 written as -0.3, plausibility overriding the data it had just read), and a pipeline's quality gate did six times more damage to the pooled estimate than the planted sabotage it caught [Barra, 2026]. The harness comprises: schema-constrained typed calls (`{"function", "arguments", "source", "derivation"}`), an arity check with the full function signature, a sign-echo net, an arm-agnostic declared-source check, a declared-derivation check, an argument type system with per-class value impossibilities, a confidence-interval bound-order check, a coherence check tying each interval call to the study's own last executed mean difference, a negative-SD result warning, a closing-versus-executed check, per-call warning budgets with confirm-by-repeat, and an escalation flag (`requires_human_review`) on insistence. `gemma4:12b` then ran one formal pipeline over its own Anchor-2 sheets: per-study typed calls, a model-emitted DerSimonian–Laird pooling call over its own sextets, a product-layer coherence check on the reported intervals (flagged, never endorsed), and a narrative synthesis checked mechanically for orphan numbers. No model stage saw an answer key, a published value, or the seal; grader-side comparison happened only afterward.

2.11 Phase P5 — deployment on clean texts

The deployment phase ran the same architecture on the *original, unperturbed* texts of both anchors with no answer key in any loop — the day-to-day usage scenario. On clean texts, agreement cannot distinguish reading from recall, so this phase makes no reading-proof claim — the perturbed phases carry that burden. Its components: (i) fresh two-replicate extraction of all 21 primaries by `gemma4:12b` under the English instruments; (ii) a frozen errata-cell panel testing, at every source-confirmed erratum of Anchor 1 reachable by the sheet, whether the model's cell sides with the source or with the anchor's published (erroneous) value; (iii) deterministic recomputation of both meta-analyses and per-trial three-way tables (source quotation × the anchor authors' cell × the model's cell); and (iv) a three-configuration comparison of error detection over the same sheets — (a) the deterministic engine with no nets, (b) the engine plus deterministic sheet-versus-source nets, and (c) the model orchestrating under the conversational ten-net harness — each registered as a dated amendment before its run.

2.12 Comparison conventions and software

All computation used Python 3.12 with the repository's validated engine functions (risk ratio and its log-scale interval; mean difference and its interval; SD conversions; Mantel–Haenszel and DerSimonian–Laird pooling; inverse-variance pooling), each frozen before the campaign; the risk-ratio function reproduces a published per-study value of Anchor 1 as its standing test case. No external statistics package was used. Replicates measure stability, not statistical significance, and no formal between-model inference was attempted: with one corpus per anchor the model comparisons are descriptive, and every superlative in this article is scoped to these corpora. The campaign's compute totalled 277 model calls across P1–P4 (≈10.8 h of machine time), with every queue resume-safe and every phase committed to the public repository before it ran.

2.13 Robustness analysis: instruction language

Because the instruments were drafted in Portuguese before being frozen in English, and the instrument-development phase had exercised the identical corpora, seals, keys, engines and graders under the Portuguese drafts, that archived record provides a natural comparison arm in which instruction language is the only changed variable. For each phase, the directly comparable quantities of the two records were compared, with the pre-registered acceptance criterion that differences stay within replicate-level variance.

3 Results

3.1 P1 — extraction: the reading proof held, and no attributable recitation occurred

All 140 extraction calls (5 models × 14 trials × 2 replicates) completed with clean stops in 390.4 minutes, and every sheet parsed at the sheet level (one single replicate, `deepseek-r1:14b` on one trial, was unparseable; its other replicate served). Cell agreement with the corrected two-layer key ranged from 88/124 (71.0%) for `llama3.1:8b` to 104/124 (83.9%) for `qwen3:14b` and `qwen3.5:9b`, with `gemma4:12b` at 103/124 (83.1%); replicate stability ranged from 77.4% to 99.2% (Table 2). Because the numeric comparator cannot see format or granularity equivalence, cross-model ranking by this score alone is provisional; the value-level separation appears in P3. The point matters concretely: `qwen3:14b`'s one-cell edge over `gemma4:12b` here dissolves under the factuality decomposition below (six omissions against zero — and omissions are real misses where format variants are not) and reverses decisively at the value level in P3.

The mechanical recitation detector raised 5 candidate cells across 140 sheets, and adjudication against the perturbed source texts resolved all 5 to incomplete-perturbation artifacts rather than recall: three models independently wrote a displaced count (113) that is arithmetically reconstructible from a surviving percentage printed beside it in the perturbed text (57.7% × n=196 = 113.1), and two models transcribed a value (4088) that survives verbatim in a prose restatement the perturbation operator missed. Zero attributable recitations remained. The same three models copied, rather than reconstructed, the equally inconsistent control-arm cell of the same table — an asymmetric preference shared across architectures, recorded as a behavior note.

The factuality classification of the divergent cells (21–36 per model) found **zero invention candidates in every model**: each number each model wrote exists in, or is arithmetically derivable from, the perturbed text it read. Omissions accounted for 0 cells in `gemma4:12b` and `llama3.1:8b`, 2 in `deepseek-r1:14b`, 4 in `qwen3.5:9b` and 6 in `qwen3:14b`; the remainder were value or format divergences. Most divergences were themselves replicate-stable — the same non-key answer written in both independent readings: 17/21 for `gemma4:12b`, 20/20 for `qwen3:14b`, 13/20 for `qwen3.5:9b`, 16/26 for `deepseek-r1:14b`, but only 17/36 for `llama3.1:8b` — suggesting that for the stronger readers, divergence from the key is a systematic reading choice rather than sampling noise.

Table 2: P1 extraction under English instruments (perturbed Anchor-1 corpus; 124 eligible key cells per model). The two scores use different comparators: *cells vs key* grades replicate 1 against the source-verified key (accuracy), while *replicate stability* grades replicate 1 against replicate 2 — the model against itself, no key involved (reproducibility). A cell can be stable yet divergent from the key, so stability can exceed the key score. Recitation candidates are shown before/after adjudication.

Model	Cells vs key	Replicate stability	Recitation cand.	Parse failures
gemma4:12b	103/124 (83.1%)	119/124 (96.0%)	1 → 0	0
qwen3:14b	104/124 (83.9%)	123/124 (99.2%)	1 → 0	0
qwen3.5:9b	104/124 (83.9%)	103/124 (83.1%)	0	0
deepseek-r1:14b	97/124 (78.2%)	96/118 (81.4%)	1 → 0	1 replicate
llama3.1:8b	88/124 (71.0%)	96/124 (77.4%)	2 → 0	0

3.2 P2 — arithmetic: unaided fails, the calculator repairs

Across all five models, exactly 1 of 57 unaided 95% confidence intervals was exact (arm A). With the text-protocol calculator (arm B), exact intervals rose to 41/50 (82%) among the four models that operated the tool — 11/16 for `gemma4:12b`, 14/16 for `qwen3:14b`, 8/9 for `llama3.1:8b`, 8/9 for `qwen3.5:9b` — and exact point estimates rose in step (4→12, 4→14, 5→8, 5→8). Pooled estimates remained at or near zero exact in both arms for every model in the cast, with or without the tool. `deepseek-r1:14b`'s unaided arm was unusable (5 of 6 outputs unparseable, zero gradable quantities), and its pooled-family tool arm ran without emitting a single `CALC:` line.

3.3 P3 — creation: every model reproduces the dichotomous pools; exactly one reaches the continuous diamond

From the P1 sheets, the deterministic engines reproduced Anchor 1's published pooled estimates for *all five* models: pooled morbidity 0.771–0.779 against the published 0.778 [0.567, 1.068], and pooled mortality 0.980–1.023 against the published 1.021 [0.446, 2.337], every value within ±0.01 of the published except one mortality estimate at Δ0.04 (Table 3). The postoperative-ileus pool is non-comparable by construction (anchor erratum #16: one published row is a different outcome) and was reported, never pooled against: over the two genuinely-ileus trials, the pool was 0.821 [0.559, 1.207] under Mantel–Haenszel and 0.429 [0.048, 3.87] under DerSimonian–Laird — both crossing 1 — identically from the perturbed and the clean-text extractions.

Table 3: P3, Anchor 1: DerSimonian–Laird pools from each model's own sheets (published: morbidity 0.778 [0.567, 1.068]; mortality 1.021 [0.446, 2.337]). Every model contributed every trial of the anchor's own per-outcome roster (4/4 for morbidity, 2/2 for mortality — a meta-analysis pools, per outcome, only the trials reporting it; Section 2.2), so no trials-in-pool column is needed here — unlike Anchor 2 (Table 4), where sheet starvation varies by model.

Model	Morbidity RR [95% CI]	Mortality RR [95% CI]
gemma4:12b	0.779 [0.569, 1.065]	1.023 [0.447, 2.344]
qwen3:14b	0.779 [0.569, 1.065]	1.021 [0.446, 2.337]
llama3.1:8b	0.771 [0.562, 1.059]	1.023 [0.447, 2.344]
qwen3.5:9b	0.779 [0.569, 1.065]	1.019 [0.445, 2.336]
deepseek-r1:14b	0.775 [0.565, 1.063]	0.980 [0.428, 2.246]

From the fresh Anchor-2 extractions (70/70 calls, 155.1 minutes), the same engine and sealed lens produced five diamonds (Table 4). `gemma4:12b`'s lens landed at −0.27 [−0.38, −0.17] (I^2 28.6%), beside the published −0.24 [−0.32, −0.16]. The other four landed between −0.47 and −0.80 with saturated heterogeneity (I^2 90.5–99.1%), two of them with starved sheets (5/7 and 6/7 trials in the pool). The invention screen over all 1,104 Anchor-2 sheet cells (five models, both replicates) found zero invention candidates in any model.

Table 4: P3, Anchor 2: the five diamonds — unperturbation-lens pooled HbA1c mean differences (published: −0.24 [−0.32, −0.16], I^2 6%).

Model	Lens MD [95% CI]	I^2	Trials in pool
gemma4:12b	−0.27 [−0.38, −0.17]	28.6%	7/7
deepseek-r1:14b	−0.47 [−1.22, 0.28]	99.1%	7/7
llama3.1:8b	−0.60 [−1.02, −0.18]	90.5%	5/7
qwen3:14b	−0.63 [−1.08, −0.17]	95.2%	6/7
qwen3.5:9b	−0.80 [−1.67, 0.07]	95.0%	6/7

3.4 P4 — orchestration under the English harness

`gemma4:12b` closed all 7 per-study call sequences under the English ten-net harness in 8.7 minutes. Two content warnings fired — both on one trial, both from the declared-derivation net — and both were resolved by the confirm-or-correct mechanism; no other net fired and no format failure occurred. The model then emitted the pooling call over its own seven sextets and closed digit-consistently with the executed result (−0.50 [−0.78, −0.22]); the mechanical truth over the same sheets was −0.52 [−0.82, −0.22] (ΔMD 0.02, from two named route divergences). The product layer flagged zero incoherent intervals, and the synthesis contained zero orphan numbers. The grader-side lens over the same sheets was −0.27 [−0.38, −0.17].

3.5 P5 — deployment on clean texts: the errata are visible, and the one miss is one cell

On the original, unperturbed texts, replicate agreement was 96.8% (Anchor 1) and 95.4% (Anchor 2), and all 42 sheets parsed. At every cell of the frozen errata panel — 7 of 7, including all four direction-critical swaps — the model's extraction sided with the *source* against the anchor's published value; on the trial whose published ileus row reproduces its pulmonary-complication counts (erratum #16), the model wrote *not reported* in both replicates rather than reciting the published numbers. The deterministic recomputation reproduced Anchor 1's pooled morbidity (0.779 [0.569, 1.065]) and mortality (1.019 [0.445, 2.336]) within tolerance; the Anchor-2 pool missed its tolerance at −0.34 [−0.51, −0.18] against the published −0.24 [−0.32, −0.16], and the residue decomposed entirely to one cell — the model transcribed a printed "−1.6 ± 0.3" from one trial's results prose and declared the ± a standard deviation, where the value equals the half-width of the confidence interval printed in the same trial's table. Correcting that single cell by the key's own conversion rule yields −0.24 [−0.32, −0.16] (I^2 3.7%), the published diamond to the hundredth; the cell was identical in both replicates, and zero invented values occurred in the phase's 124 graded cells.

Over the same sheets, the three detection configurations behaved differently against that poisoned cell (Table 5): the bare engine delivered the distorted pool silently; the sheet-versus-source net flagged the experimental arm mechanically, with both source quotations, after a first rule revision (its initial form stayed correctly silent because the trial's own prose prints the ± form — the flag rule was extended, as a dated amendment, to fire when a printed ± equals the printed interval's half-width; the control arm's ± matches nothing within the pre-registered tolerance and was not flagged); and the conversational ten-net harness raised zero flags on the poisoned cell — whose downstream calls were well-formed, correctly source-declared and internally coherent — while catching four live orchestration slips (one arity, three derivation declarations) in the same run. A DerSimonian–Laird weight-share flag (>40%) was blind to the distortion (the poisoned trial's share was 19.3%; the distortion's product signature was heterogeneity inflation, I^2 6%→79%).

Table 5: P5: three error-detection configurations against the same single-cell defect, same sheets, no answer key in any loop.

Configuration	Pool delivered	The poisoned cell
(a) deterministic engine, no nets	−0.34 [−0.51, −0.18]	silent
(b) engine + sheet-vs-source nets	−0.34 (unchanged; warn-only)	flagged (exp. arm, with quotes)
(c) model + conversational ten-net harness	−0.32 [−0.46, −0.19]	not flagged; 4 live slips caught

3.6 Instruction-language robustness

Per phase, the campaign's results sat within replicate-level variance of the archived instrument-development record, in which identical materials had run under the Portuguese instrument drafts (Table 6). The largest reading-stage difference was +3 cells of 124 on the one directly comparable model, with replicate stability identical to the decimal; the arithmetic signature (unaided failure, calculator repair, pool ceiling, per-model deviant classes) matched point for point; the creation-stage flagship lens was identical in point estimate *and* interval (−0.27 [−0.38, −0.17] in both records); and the orchestration-stage delta sat inside the measured route-variability band with the same warning classes on the same trial.

Table 6: Instruction-language robustness: the campaign (English instruments) vs the archived instrument-development record (Portuguese drafts), per phase (directly comparable quantities only).

Phase	Campaign (English)	Development record (Portuguese)
Read (gemma4:12b, cells / stability)	83.1% / 96.0%	80.6% / 96.0%
Calculate (unaided exact CIs; tool arm)	1/57; 82% exact	0/30; same signature
Create (gemma4:12b lens, Anchor 2)	−0.27 [−0.38, −0.17]	−0.27 [−0.38, −0.17]
Orchestrate (pool vs truth; behavior)	Δ0.02; confirm-or-correct	Δ0.06; same classes

4 Discussion

This study measured, in one continuous pre-registered chain, what small local models running on an integrated GPU can contribute to meta-analysis — and where the risk concentrates. The models read clinical trials reliably enough to rebuild two published meta-analyses; they cannot do the interval arithmetic unaided but do it almost perfectly through a trivial text-protocol tool; a deterministic architecture turns their sheets into pooled estimates that match the published ones; and exactly one model in the class produces a continuous-outcome diamond beside the published value. Under the perturbation control, none of this is attributable to memory of the published review.

The five-model comparison exposes a clean division of difficulty, and the mechanism appears to be structural as much as cognitive. On the dichotomous side, the raw material is transcription — integer counts and denominators printed redundantly across table, prose and percentage

— and the pipeline then *dampens* what error remains: a risk ratio is a ratio, so choosing a different population layer shifts numerator and denominator together, and the pooling weights derive from the counts themselves. The evidence is in Table 3: the five models took visibly different denominator routes and still landed at 0.771–0.779 against the published 0.778. On the continuous side each sextet quantity is printed once, in one of several competing layers, and the decisive step is semantic rather than transcriptive — identifying what a printed dispersion *is*; a mislabeled type sends the conversion down the wrong route, and inverse-variance pooling then *amplifies* the error, because weight scales with $1/\mathrm{SD}^2$ — which is how one dispersion fourfold too small dominated a pool and inflated I^2 from 6% to 79% in the deployment phase. The same architecture is therefore error-dampening for count outcomes and error-amplifying for continuous ones. The five diamonds spread from −0.27 to −0.80 entirely through reading-class differences, never arithmetic — the same engine computed all five — supporting the study's central design thesis: once everything deterministic is code, reading is the whole game, and model choice is a reading choice. Even the best diamond, however, is not a full reproduction of the anchor's homogeneity: `gemma4:12b`'s lens carries I^2 28.6% against the published 6% — the same pooled effect, direction and significance over a visibly noisier reconstructed evidence body, with the residual heterogeneity plausibly injected by route-level extraction differences (population layers, dispersion conversions) rather than by any single wrong cell. "Beside the published value" is therefore a claim about the estimate, not digit-level equivalence of the whole meta-analysis — an honest bound worth stating, since the wider interval is exactly what a practitioner would inherit. Notably, none of those reading failures was a fabrication: across both corpora and all five models, the invention screen found no cell whose numbers cannot be traced to the text read, and the failure inventory consists of omissions, population-layer choices, format variants and mislabeled quantities — most of them reproduced identically across replicates in the stronger readers, i.e. systematic and therefore auditable, not stochastic. This honesty profile is what makes a warn-only architecture viable at all — detection nets suffice for models whose errors are traceable, whereas fabricating models would demand the value-substituting interventions the doctrine forbids. The pooled-arithmetic ceiling points the same way: even with the calculator, assembling the list-valued pooling calls stayed beyond every model in this size class (in instrument-development testing, only a 27-billion-parameter model ever mastered that call) — which is precisely why the architecture moves pooling into code rather than teaching it. For practitioners this suggests a concrete triage: dichotomous evidence tables may be delegated broadly, while continuous outcomes concentrate all the model risk and deserve the strongest reader plus the strongest checks.

The deployment phase turns the benchmark toward real use, where no answer key exists. Its clean-text results support the day-to-day scenario in three ways. First, the model's extractions sided with the primary sources at all seven source-confirmed errata of the published meta-analysis — including refusing, in both replicates, to supply the ileus counts that the anchor publishes but its source never reports — so a local reader plus a three-way table (source quotation, human cell, model cell) makes published extraction errors *visible* to a reader rather than requiring trust in either extractor. One of those errors carries conclusion-level weight (Table 1, #16): the anchor's significant ileus reduction (abstract RR 0.48) rests on the mislabeled pulmonary-complication row — over the two genuinely-ileus trials the pool crosses 1 under either method (0.821 [0.559, 1.207] MH; 0.429 [0.048, 3.87] DL) — while its no-difference conclusions for morbidity and mortality survive every documented erratum, as the five-model reproductions demonstrate. This is consistent with reports that language-model assistance can match human extraction quality in live reviews [Lee et al., 2026], and it adds the complementary observation that the disagreements are informative in both directions: the same source-verification process also corrected the benchmark's own key twice during development, and the errata file carries the adjudicators' own published errors. Second, the phase's one failed tolerance decomposed to a single cell with an instructive anatomy: the primary itself prints the same quantity in two layers — a table interval and a prose "±" that equals the interval's half-width — and the model transcribed the prose layer and mislabeled the ± as a standard deviation, identically in both replicates. The defect is seeded by the source's own ambiguous notation, and it defines this architecture's characteristic residual failure class: a model that misdeclares its sheet escapes every check that trusts the sheet. The instrument asymmetry likely mattered here: the continuous sheet carries no provenance field, so the sheet could not say *which* printed layer the 0.3 came from — on the dichotomous sheet, whose mandatory `where` locator repeatedly disambiguated population layers during adjudication, the prose-versus-table conflict would have surfaced at extraction; extending provenance (or short verbatim-quote) fields to every sheet is the corresponding instrument work, queued for future versions and never retroactive. Third, the three-configuration comparison shows that detection layers are complements, not substitutes: conversational nets catch conversation-layer defects (four live orchestration slips in a single run) but are structurally blind to a well-formed poisoned cell, which only a sheet-versus-source net saw — and a weight-share product flag was blind too, because random-effects weighting re-equalizes shares while the distortion surfaces as heterogeneity inflation. No warn-only configuration made the wrong number right — by doctrine it must not — so the safest key-free architecture this record supports is a deterministic engine, sheet-versus-source detection nets, a conversational harness for orchestration slips, and a reviewer dispatched to the primary's table whenever any flag fires — a role played in this study by the human–AI adjudication pair, whose flag resolutions are themselves quotation-bound and on record.

These results sit within a fast-moving literature. Feasibility studies report promising but uneven extraction accuracy for cloud models [Schmidt et al., 2024]; screening assistance is approaching routine [Xie et al., 2026]; and end-to-end platforms report complete reviews in hours [Bakker et al., 2026, McLaughlin et al., 2026]. In contrast to that line, everything measured here runs locally, at zero marginal cost, on hardware compatible with the privacy constraints of clinical text [Wiest et al., 2024, Spaanderman et al., 2025] — and under evidence-synthesis organizations' stated conditions of demonstrated rigor and transparent reporting [Flemyng et al., 2025], which the perturbation proof, the public seals and the adjudication rite are designed to operationalize. The reading-versus-recall control matters more, not less, as anchors and primaries enter training corpora [Xu et al., 2024]: in 140 perturbed sheets the campaign found zero attributable recitations, while both mechanical candidates it did raise traced to survivable gaps of the perturbation operator itself — now demonstrated across multiple architectures and queued as instrument fixes.

A thread that runs through both the instruments and the findings is plausibility bias — the tendency of any judge, human, hybrid or model, to prefer a plausible reading over the printed one. It shaped this study twice before a single campaign run: the adjudication rite exists because the benchmark's own judge was reversed six times during key construction, twice while wrongly accusing models of fabrication, and the warn-only doctrine exists because development-phase checkers with write-power repaired correct values into wrong ones. In the campaign itself the bias was then measured, not presumed — and it appeared exactly once, on the judge's side: the deployment phase's record initially charged the model with *computing* the mislabeled dispersion, until a detection-net run showed the value printed verbatim in the source's own prose, and the charge was corrected with a dated note in the archived record [Barra, 2026]. The model stages showed zero occurrences: no value was ever substituted (structurally forbidden), the sign-echo net never fired, and all five models transcribed even the sign-sensitive cells verbatim — while their overall factuality profile ran opposite to the judges' (zero fabrications; failures traceable to the text). The published meta-analysis's errors were caught by quotation rather than judgment, and the three-configuration comparison rewards architectures whose checks can only point, never overwrite. This aligns with the documented biases of model judges [Zheng et al., 2023] and suggests they generalize beyond model judges:

the constraint that works is structural — quote before verdict, flag before fix — not a better judge.

The instruction-language robustness check closes a confounder rather than opening a finding. Because the instruments were drafted in Portuguese and frozen in English over otherwise identical materials, a referee could ask whether either language handicapped — or peculiarly helped — models trained predominantly on English; the comparison answers by measurement. Per phase the two records agree within replicate-level variance — the one directly comparable reading score moved three cells in 124, the arithmetic signature matched point for point, and the flagship diamond was -0.27 [-0.38, -0.17] in both, interval included — which suggests that multilingual instruction-following is effectively solved at these model sizes for structured tasks, and establishes that none of the conclusions above is an artifact of instruction language.

Several limitations bound these claims. Two anchors from a single journal family and 21 primaries are the whole evidence base; every superlative is scoped to these corpora, and no formal between-model inference is attempted. One quantization per model and one hardware profile were tested. The mechanical cell comparator is a declared approximation of the full adjudicated ruler, so absolute cell scores understate agreement and cross-model gaps partly reflect format habits. The perturbation operator's residual gaps (derivable totals; prose restatements) are documented and exploitable, though they produce reconstruction, not recall. The English harness build reproduces the frozen net logic but was exercised in one formal run of one model. The adjudicator is the same human–AI pair that built the harness, mitigated by mandatory quotation and public self-errata rather than independence, and the known biases of model judges [Zheng et al., 2023] are constrained, not eliminated, by that rite. Finally, the deployment phase's clean texts cannot separate reading from recall by design; its claims rest on the perturbed phases' proof.

In conclusion, a consumer machine running one well-chosen 12-billion-parameter local model, a frozen deterministic engine and warn-only detection nets can rebuild published meta-analyses to the published digits, expose documented errors in them with source quotations, and fail in ways that are named, single-celled and flaggable — and none of this depends on the instruction language. The designed next step is the generalization gate: the same frozen architecture against a new anchor in a new clinical domain, where transfer, not replication, is the question.

Data and code availability

All protocols (pre-registered, with dated amendments committed before each run), the frozen English instrument library with its Portuguese-correspondence tables, perturbation seals with SHA-256 fingerprints, the two-layer answer keys and the anchor errata file, the deterministic engines and graders, the English harness build, all 277 raw model outputs and run logs, per-phase evaluation records, adjudications with source quotations — consolidated in a raw-data index of the adjudication layer (`dados/adjudication-record.md`: the quotation-bearing key cells, the judge's reversals and withdrawn accusations, the campaign-era correction with its refuting net run, and every flag resolution) — and the exact quantized builds (pinned by Ollama digest) are public at `https://github.com/gbbarra/llm-evidence-extraction-benchmark-ptbr` and archived at Zenodo [Barra, 2026]. Closed-stratum primary articles are not redistributed; scripts rebuild that corpus stratum from legally obtained copies.

Declarations

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